



Clinical Practice Guideline

Wound Management

Reference #: NPG16

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to Wound Management. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

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Principles

Observe the five moments for hand hygiene. Follow aseptic technique when performing wound care and use the risk assessment outlined in the SCGOPHCG Infection Prevention and Management Manual- Aseptic Technique Policy. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.

Adhere to the requirements for medical photography as outlined in the SCGOPHCG Clinical Photography and Audio-Visual Recordings Policy.

SCGOPHCG recognises the role of carers and shall establish and maintain mechanisms for the involvement of carers in the provision of services

1.0 General Management 6,8,9,10,12,14,16,22,25,27

1.1 Medical Requirements

Certain topical products require prescription (e.g. Silver Sulphadiazine cream (SSD), Metronidazole gel).

1.2 Safety Information

Assessment of radial artery patency and collateral competency should be undertaken prior Hydrogen Peroxide is not recommended to cleanse or irrigate cavities due to potential risk of air embolism.

1.3 Nursing Management

If medical photography is required staff must adhere to the SCGOPHCG Clinical Photography and Audio-Visual Recordings Policy.

OPH: utilise the Silhouette Star system (camera located at ward 4 nurse's station).

All patients require a full skin and wound assessment to be completed on admission

Note: OPH Rehabilitation and Aged Care (RAC) wards must complete within 2 hours of admission, all other patients complete within 8 hours of admission.

Documentation on the Wound Management Plan must include:

- wound type,
- location,
- dimensions,
- appearance,
- exudate.

The Wound Management Plan should be discussed with the patient / carer and documented in the inpatient notes.

Patients wound and dressing management including outcome of all assessments, interventions and specific site wound management is to be accurately and legibly recorded in the patients' health records.

Assessment of pain specific to the patients wound should be performed with each dressing change. Consider administration of analgesia as indicated / prescribed. Recommend referral to Acute Pain Service as indicated.

Consider the following factors that impair wound healing (Table 1):²¹

Table 1: Factors that may impair wound healing

Intrinsic factors	Extrinsic factors	Local factors
• Malnutrition Obesity • Psycho-social well-being • Diabetes • Arterial disease PVD • Advanced age • Disease Process	• Smoking, alcohol abuse • Medications • Radiotherapy, Chemotherapy • Temperature	• Desiccation & Maceration of the wound bed, • Infection, • Pressure, shear and friction, • Foreign body, • Dressing selection, • Chemical stress, • Wound pain

Wound Cleansing

The primary objective of wound cleansing is to remove debris, previous dressing material, foreign bodies from the wound bed.⁴

Normal saline is the recommended cleansing agent to be used.

Acute Wounds

- Acute wounds are surgical or traumatic wounds that heal in an orderly timely manner within 4- 6-week period ²¹.

Chronic Non-Healing Wounds (e.g., leg ulcer) ⁸

- Potable tap water is suitable for adults with chronic wounds
- When using a clean wound management technique (e.g., Immersion of the limb in a bowl /bucket lined with disposable plastic bag) potable tap water or normal saline may be used for irrigation.

1.4 Procedural Information**Requirements**

- Wet strength bag
- Dressing pack
- Normal Saline
- Sterile instruments and dressings as required
- Personal Protective Equipment (PPE)

Procedure

1. Administer analgesia if required.
2. Perform hand hygiene.
3. Gather equipment for the procedure.
4. Perform hand hygiene and put on non-sterile gloves.
5. Cleanse wound with warm tap water +/- a soap free pH balanced skin cleanser if available.
6. Dry the wound using a clean towel.
7. Remove gloves and perform hand hygiene
8. Clean the trolley with a combined detergent/ disinfectant wipe.
9. Gather equipment for the procedure.
10. Perform hand hygiene.
11. Open dressing pack using the corners and add items required using a non-touch technique.
12. Perform hand hygiene and put on non-sterile gloves.
13. Document any change in condition of wound.

Documentation

- The patients wound and dressing management including outcome of all assessments, interventions and specific site wound management is to be accurately and legibly recorded in the patients' health records.
- Patient education should be given in the form of verbal and written.

Discharge Planning

- Where patients require a continuation of management of their wound referral to the appropriate clinical services on discharge should be performed.
- Ensure referral and current wound management plan are supplied to the service provider
- Provide education and a current wound management plan to the patient and or carer
- Provide additional wound supplies on discharge
- Ensure the patient/carers are aware of any follow-up appointments and/or service referrals
- Provide education to the patient/carers regarding adverse signs and symptoms:
 - Infection
 - Wound deterioration, or other potential problems, e.g. pain or bleeding.

2.0 Skin Tear Prevention and Management ^{12, 13}**2.1 Medical Requirements**

Medical team review of patient immunisation status is required.

2.2 Safety Information

Only soft silicone dressings should be used as a primary dressing on skin tears e.g., Mepitel or Mepilex

The following adhesive dressings and tapes should not be used on those who are susceptible to skin tears due to fragility and high risk of further skin injury (see procedural information).

- Fixation tapes (e.g., Fixamull), skin closure (e.g., steri strip)
- Film dressings (e.g. Tegaderm)
- Hydrocolloid dressings (e.g. Duoderm)
- Adhesive Tapes or adherent dressings that may adhere and cause trauma.

2.3 Nursing Management

Use the STAR Skin Tear Classification System (please refer to Table 2, page 8) to categorise the type of Skin Tear ¹². A skin tear can be:

- partial thickness (separation of the epidermis from the dermis), or
- full thickness (separation of both the epidermis from the dermis underlying structures)
- All large full thickness skin tears must be reported to the admitting medical team and referred to Plastic's team for assessment and management.

Any skin tear acquired via iatrogenic causes must be reported using the Datix Clinical Incident Management System (CIMS).

Patients at risk of skin tear development should have appropriate interventions implemented.

Adhesive skin closures (steri-strips) should not be used at any time on Skin Tears, as this creates traction/torsion of skin and underlying tissue. Suturing and stapling of skin flap are not current best practice.

Risk assessment

- In addition to assessment on admission, patients should be assessed on at each episode of care to identify their risk of sustaining a skin injury.
- Perform a comprehensive head to toe skin assessment using skin characteristic guide on the Pressure Injury Risk & Skin Integrity Management Plan.

If one or more of factors listed below are present; patient is at risk of sustaining a skin tear⁷: •Extremes of age •Fragile skin •Previous skin tears General Health •Chronic/critical disease •Polypharmacy •Impairment: cognitive, sensory, visual, auditory, nutritional Mobility •History of falls •Impaired mobility •Dependent activities of daily living •Mechanical trauma	Patients increase to high risk of sustaining a skin tear in the hospital environment if any of the factors listed below are present.⁷ •Decreased activity, mobility or sensory changes related to changes in health status / medications •Skin dry with presence of ecchymosis – bruising and discoloration of skin •Nutritional status is > 2 on Malnutrition Screening Tool •Dehydration and decreased skin turgor •At risk of pressure /friction /shearing of their skin. •Waterlow Score >10 (SCGH) •Braden score < 18 (OPH) •Patient that is cognitive impaired and in an unfamiliar environment
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Strategies to prevent Skin Tears

- Those patients at risk should have standard interventions implemented as per Pressure Injury Risk & Skin Integrity Management Plan as well as patient specific interventions – See table 1.

Table 1: Preventative Strategies

Appropriate Skin Care ^{8 level 1}	- Use natural pH skin Cleansers –avoid soap - Encourage BD skin moisturising with a neutral PH moisturiser
Safe Environment 8 level 5	- Encourage clothes, which are not restrictive, loose long sleeves and long trousers. - Ensure safe environment – no equipment with sharp edges. Foam padding may be required. - Encourage patients to wear their glasses, hearing aids and appropriate non-slip footwear. - Ensure clinical staff and patients avoid wearing jewellery, watches and maintain short nails
Injury Prevention ^{8 level 5}	- Encourage patient mobilisation with care – communication and plan prior to patient movement. - Correct manual handling technique to prevent friction and shear when turning, lifting and transferring. Use of slide sheets, hover mat etc. - Prevent falls injuries. Undertake mobility and falls risk assessment and implement falls risk strategies. - Protect patients' skin and limbs from skin injury – consider extra padding of limbs e.g. foam held insitu with tubular band - Provide patient education to patients regarding promoting healthy skin e.g. ceasing smoking, preventing friction and shear - Discuss cause and risk of sustaining a skin injury with your patient/ family/carers and prevention and management plan.
Clinical Considerations	- Optimise nutrition and hydration. Refer to Dietician When Nutritional score >2 on MST - Review long-term use of medications known to alter skin integrity e.g. long-term use of steroid ointments. - Adhesive dressings and tapes not to be used at all. - Use of soft silicone tapes e.g. 3M Kind remove - Soft silicone wound products only to be used when dressing skin tears e.g. Mepitel or Mepilex foam border. - Prophylactic dressing to prevent shearing and friction over high-risk areas

Presence of Skin Tear

- Mark the Skin Tear (ST) on the diagram on the front of the Wound Management Plan
- Skin Tear Category should be classified using the Star Skin Tear Classification guide (see table 2).
- Assessment of skin tear wound appearance should be documented in the integrated notes:
 - Location of the skin tear
 - Size of skin tear Small < 5x5cm, Medium > 5x5cm, Large > 10x10cm
 - Presence of skin flap and appearance.
 - Identify if Skin Tear is Partial thickness or full thickness.
 - Presence of Pain.

- Exudate amount, type and consistency.
- Peri-wound skin appearance.
- Initiate a Wound Management Plan
- Skin Tear acquired in hospital may be reported using Clinical Incident Management System.
- Ensure communication with interdisciplinary health team so they are aware of skin integrity issues identified and prevention/ treatment strategies in place.
- Medical review is required of full thickness and large skin tear and subsequent referral to Plastic's Registrar.
- Referral to Wound Management Nurse is required for those sustaining large skin tears or at any time wound management advice is required.
- Monitor and record the effect of interventions implemented.
- Commence and document daily skin and Waterlow risk assessments.
- The patients wound and dressing management including outcome of all assessments, interventions and specific site wound management is to be accurately and legibly recorded in the patients' health record.

2.4 Procedural Information

Requirements

- Dressing pack
- Wet Strength Bag
- Personal protective equipment (PPE)
- 0.9% Normal Saline
- 10ml syringe.
- Sterile Cotton buds/ additional sterile gauze if large skin tear.
- Disposable sterile metal forceps/scissors
- Appropriate dressing of choice following wound assessment
- Primary dressing soft silicone dressing e.g., Mepitel – silicone tulle dressing or Mepilex foam with border
- Secondary dressing e.g., pad
- Extra padding e.g., foam or pad, tubular band or under padding and crepe bandage.

Procedure

Aseptic Technique Wound Care (Uncomplicated)

1. Administer analgesia if required.
2. Perform hand hygiene.
3. Clean the trolley with a combined detergent/ disinfectant wipe.
4. Gather equipment for the procedure.
5. Perform hand hygiene.
6. Open dressing pack using the corners and add items required using a non-touch technique.
7. Perform hand hygiene and put on non-sterile gloves.
8. Remove soiled dressing using a non-touch technique, (If the dressing is adhered then irrigating with saline in a 10ml syringe may be required to assist with removal) and discard. Remove gloves.
9. Perform hand hygiene and put on non-sterile gloves.
10. Cleanse the skin tear with gauze and normal saline. Using forceps remove any debris or residual haematoma and gently irrigating with normal saline in a syringe to ensure area thoroughly cleansed. Disposable metal forceps maybe required to do this.
11. Assess Skin Tear size, wound and degree of tissue and skin loss. Identify presence and colour of any skin flap. Use the Star Skin Tear Classification System (refer to Table 2, page 8) to categorise the type of Skin Tear.
12. If a skin flap is present (category 1A, 1B, 2A and 2B) then realign skin flap using fine

metal forceps and/or normal saline moistened cotton buds to gently roll oppose skin edges where possible. Do not cut off skin flap.

13. Redress the wound as per nursing care plan below

Small Skin Tears that are not bleeding or do not have large exudate	Moderate to large size skin tears or a skin tear that is bleeding or has a large exudate
• Cover Skin Tear with soft silicon foam e.g., mepilex foam border, which will secure the skin flap in position. Ensure there is an overlap of at least 3cm onto the surrounding healthy skin. • Mark the skin flap position and size on the outside of the dressing. • Draw an arrow to indicate removal direction so no trauma is caused to healing flap when dressing removal. • Aim to leave initial dressing on for 3-days unless exudate strikes through dressing increase in pain at site or surrounding erythema.	• Primary dressing - Stabilise and secure skin flap by applying a fenestrated sheet of soft silicon mesh e.g., Mepitel over the skin tear. This is a non-adherent silicon dressing and secures the wound edges holding the skin flap in position. • The primary dressing of silicon mesh needs to overlap onto healthy peri wound skin by at least 3cm around entire wound and may remain insitu for 5 days or longer if requested by Wound Management Nurse or Plastic's team. • If wound still bleeding apply a calcium alginate dressing over the primary dressing of silicon mesh to assist with haemostasis e.g., Curasorb. Secondary dressing – cover with an absorbent pad. • Hold dressing in situ with underpadding e.g., kerlix and / or crepe bandage.

14. Dispose of waste.

15. Remove gloves if used, perform hand hygiene.

16. Clean the trolley with a combined detergent/ disinfectant wipe.

17. Perform hand hygiene.

18. Document findings and assessments on wound management plan

Discharge Planning

Where patients require a continuation of management of their skin tear referral to the appropriate clinical services on discharge should be performed.

- Ensure referral and current wound management plan are supplied to the service provider
- Provide education and a current wound management plan to the patient and or carer
- Provide additional wound supplies on discharge
- Ensure the patient/carers are aware of any follow-up appointments and/or service referrals
- Provide written and verbal education to the patient/carers regarding adverse signs and symptoms:
 - Infection
 - Wound deterioration
 - Other potential problems, e.g., pain or bleeding



STAR Skin Tear Classification System Guidelines





Government of Western Australia
Department of Health
North Metropolitan Health Service



1. Control bleeding and clean the wound according to protocol.
2. Realign (if possible) any skin or flap.
3. Assess degree of tissue loss and skin or flap colour using the STAR Classification System.
4. Assess the surrounding skin condition for fragility, swelling, discolouration or bruising.
5. Assess the person, their wound and their healing environment as per protocol.
6. If skin or flap colour is pale, dusky or darkened reassess in 24-48 hours or at the first dressing change.

Category 1a
A skin tear where the edges can be realigned to the normal anatomical position (without undue stretching) and the skin or flap colour is not pale, dusky or darkened.

Category 1b
A skin tear where the edges can be realigned to the normal anatomical position (without undue stretching) and the skin or flap colour is pale, dusky or darkened.

Category 2a
A skin tear where the edges cannot be realigned to the normal anatomical position and the skin or flap colour is not pale, dusky or darkened.

Category 2b
A skin tear where the edges cannot be realigned to the normal anatomical position and the skin or flap colour is pale, dusky or darkened.

Category 3
A skin tear where the skin flap is completely absent.







STAR Classification System

References:

1 Payne, R., & Martin, M. (1993). Defining and classifying skin tears: Need for a common language ... a critique and revision of the Payne-Martin Classification system for skin tears. *Ostomy Wound Management*, 39(5), 16-20. 2 Photographs courtesy of the Skin Tear Audit Research (STAR) photographic library, Silver Chain Nursing Association and School of Nursing and Midwifery, Curtin University of Technology. 3 Carville, K., Lewin, G., Newall, N., Haskelhurst, P., Michael, R., Santamaría, N., & Roberts, P. (2007). STAR: A consensus for skin tear classification. *Primary Intention*, 15(1), 18-28.

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Table 2: STAR Skin Tear Classification ²

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3.0 Skin Closure Removal - Suture and Staple Removal ^{18, 19}

3.1 Medical Requirements

Remove sutures or staples from healed wounds as directed by medical officer.

3.2 Safety Information

Prior to removal of sutures or staples review the order for removal and note any specific instructions

Wound healing at skin surface normally occurs between 5 - 14 days Approximation and healing at wound edges should be assessed prior to ensure no superficial separation identified.

Observe aseptic technique when removing sutures; ensuring exposed suture material is not pulled through the healing wound, increasing the risk of wound infection.

Removal of alternative sutures or staples first allows vision of any dehiscence at skin surface. If no dehiscence noted then remaining sutures can be removed. If dehiscence identified cease removing skin closures and inform medical team caring for the patient

18 19

Use of adhesive skin closures – Steri-Strips may be required post removal to oppose skin edges if separation occurs.

3.3 Nursing Management

- Explain the procedure and obtain appropriate consent.
- Patient education to ensure they are aware they might experience a slight pulling or stinging sensation on removal of sutures or staples.
- Note amount and type of skin closures removed and if adhesive skin closure – steri-strips where required post removal.
- Record the appearance of the surgical incision and skin union after removal of sutures or staples.
- Ensure medical team and shift coordinator notified if there is evidence of dehiscence or superficial separation.
- Document appearance of surrounding skin, inflammation, exudate or pain.
- If ongoing wound care is required then commence a Wound Management Plan (MR 853)

3.4 Procedural Information

Requirements

- Dressing Pack
- Wet strength bag
- Sharps container
- Stitch cutter or staple remover
- Sterile metal disposable forceps may be required
- Normal Saline
- Wound closure strips - steri strips
- Island dressing if required
- Ensure Personal Protective Equipment (PPE)

Procedure

1. Perform hand hygiene
2. Clean the trolley with a combined detergent/ disinfectant wipe.
3. Gather equipment.
4. Perform hand hygiene, don non-sterile gloves.
5. Open dressing pack using the corners and add items required using a non-touch technique.

6. Perform hand hygiene and put on non-sterile gloves.
7. If no concerns identified with skin union then remove alternate sutures or staples using following techniques.

Suture removal technique	Staple removal technique
• Lift knot of suture slightly with metal forceps and raise it slightly. • Slide the curved tip of stitch cutter under knot of suture as close as to skin surface as possible. • Cut the suture • Pull suture out gently with the forceps to minimize trauma to skin. • Do not pull exposed suture through the wound • Inspect suture and site to ensure all material removed	Gently insert the lower jaw of the staple remover under the middle area of the staple. If the incision is under tension you may need to depress the skin surface down slightly so you can slide staple remover in under the staple. Squeeze handles of the staple remover together closing. The staple will bend in the middle and once both staple ends are free of skin, lift and remove staple gently. Place staples onto gauze or into sharps container.

8. Redress the wound as per nursing care plan
9. Dispose of waste.
10. Remove gloves if used, perform hand hygiene.
11. Clean the trolley with a combined detergent/ disinfectant wipe.
12. Perform hand hygiene.
13. Document any change in condition of wound.

4.0 Negative Pressure Wound Therapy (NPWT) ^{3, 16, 27}

4.1 Medical Requirements

Refer to medical staff or wound care specialist on required pressure settings and to check this prior to application. Documentation of the pressure settings should be recorded in the medical notes and on the NPWT Chart (MR871).

4.2 Safety Information

Return equipment to Hospital Equipment Service (HES) for cleaning and maintenance between patient applications.

Ensure the Negative Pressure Wound Therapy (NPWT) unit / machine is kept dry, DO NOT have machine connected to electrical power source if patient is in a wet area (e.g. Shower).

Contraindications ²⁸	
Necrotic tissue/eschar covering more than 25% of the wound	Exposed vasculature
Untreated osteomyelitis	Exposed nerves, vessels, tendon and ligament
Non-enteric or unexplored fistulae	Exposed anastomotic site
Malignancy in the wound	Exposed organs
Blind or unexplored tunnels	Haemophilia
Severe peripheral arterial disease	

Precautions / Considerations prior to application of NPWT (cont'd)	
Patients at high risk for bleeding or haemorrhage	Patients requiring: • MRI • Hyperbaric chamber • Defibrillation
Patients on anticoagulants or platelet aggregation inhibitors	
Patients with: • Friable vessels and infected blood vessels • Infected wounds • Osteomyelitis • Sharp edges in the wound (i.e. bone fragments) • Spinal cord injury (stimulation of sympathetic nervous system) • Enteric fistulas	Consider: • Patient size and weight • Use near vagus nerve (bradycardia) • Circumferential dressing application • Mode of therapy – intermittent versus continuous negative pressure

4.3 Nursing Management

- Assess the need for analgesia prior to dressing
- Wound assessment is to be documented on Wound Management Care Plan (MR 853) including appearance, dimensions, exudate amount and type.
- If photographic documentation is required, please refer to the SCGOPHCG Clinical Photography and Audio-Visual Recordings Policy, and at OPH use the Silhouette Star system.
- NPWT should be prescribed by the Surgical Registrar or Consultant or SRN proficient in application and assessment.
- The type of machine, dressing type, therapy parameters (e.g., intermittent, or continuous therapy and suction pressure in mmHg) and frequency of change is to be documented using the Negative Pressure Therapy Dressing Treatment Chart (MR871)
- Therapy should be monitored 8-hourly (i.e., suction settings, exudate in canister, dressing seal) and recorded on NPWT specific observation chart.
- Refer to product information/manufacturers guidelines for further information or contact Wound Management Service.
- Patients should receive the Negative pressure wound therapy (NPWT) Discharge advice booklet when leaving the hospital, or the PICO patient educational booklet if using a PICO system.

4.4 Procedural Information

Requirements

- Personal Protective Equipment (PPE)
- Wet strength bag
- Dressing pack
- Sterile drape pack
- Sterile gloves (2 pairs)
- 0.9% Normal Saline
- NPWT dressing either:
 - Foam including SensaTRAC™ pad and film drape, or
 - Renasys™ Gauze Dressing Kit with port
- Canister and tubing

- +/- Cohesive seals
- Protective barrier wipe/s
- Thin hydrocolloid sheet as required (e.g. Duoderm Thin TM)
- NPWT machine appropriate for patient's wound size and level of mobility (available from Hospital Equipment Service)
- Sterile scissors and / or scalpel
- Optional items: Spare film dressing (e.g. Tegaderm, not IV 3000) Y connector if multiple sites to be dressed

Procedure

1. Administer analgesia if required.
2. Perform hand hygiene.
3. Clean the trolley with combined detergent/ disinfectant wipe.
4. Gather equipment for the procedure.
5. Perform hand hygiene.
6. Open dressing pack using the corners and add items required using a non-touch technique.
7. Perform hand hygiene and put on non-sterile gloves.
8. Place sterile drape in position.
9. Clean wound base and peri-wound skin.
10. Dry peri-wound skin and apply protective barrier film (e.g. barrier wipe or spray).
11. Where a hydrocolloid skin barrier is required, apply thin hydrocolloid dressing to peri wound skin; allow 3-5cm border. Uneven skin surfaces may be prepared with mouldable hydrocolloid seal (e.g. Eakins ring TM) or hydrocolloid paste (e.g. Stomahesive paste TM).
12. a) If using a foam, cut to fit the shape of the wound (the foam should be slightly larger than the shape of the wound) and insert foam piece/s into the wound.
b) If using gauze, moisten with saline (the gauze should not be wet, squeeze excess Saline out) fluff and stuff into the wound space ensuring all wound edges are in contact.
13. Secure with film drape allowing more than a 5 cm overlap.
14. Cut a hole in the film drape:
 - a. Smith & Nephew System - 3 cm diameter hole
 - b. KCI System – 5 cm diameter hole.
15. Apply the softport pad (Smith & Nephew) or SensaTRAC TM (VAC) pad ensuring tubing placed in appropriate direction. Connect tubing to canister.
16. Fit canister to NPWT machine.
17. Ensure all tubing clamps are released.
18. Turn NPWT machine on, input settings and commence therapy.
19. Check that therapy/suction is working and that there are no leaks.
20. Use additional film to seal any detected leaks.
21. Dispose of waste.
22. Remove gloves. Perform hand hygiene.
23. Clean the trolley with a combined detergent/ disinfectant wipe.
24. Perform hand hygiene.
25. Document the therapy settings including the number of pieces of foam or gauze inserted into the wound, to ensure all dressing is removed at time of next dressing change.

Removal of NPWT Dressing Requirements

- Wet strength bag
- Dressing pack
- Personal Protective Equipment
- Adhesive remover wipes
- Optional (if dressing adherence):
- Sterile saline for injection
- Blunt drawing up needle
- 10 ml syringe
- Alcohol swab
- Sterile saline for irrigation

Procedure

1. Administer analgesia if required.
2. Perform hand hygiene.
3. Turn off therapy and close tubing clamps.
4. Optional: inject existing foam dressing with sterile saline solution to aid removal.
5. Wait at least 30 minutes before attempting to remove dressing.
6. Perform hand hygiene.
7. Gather equipment for the procedure.
8. Clean the trolley with a combined detergent/ disinfectant wipe. Allow to dry
9. Perform hand hygiene.
10. Open dressing pack using the corners and add items required using a non-touch technique.
11. Perform hand hygiene and put on sterile gloves.
12. Remove soiled dressing and discard.
13. Remove gloves.
14. Perform hand hygiene.
15. Clean the trolley with a combined detergent/ disinfectant wipe.
16. Document the number of pieces of foam or gauze removed.
17. Reassess wound and apply appropriate dressing.

Changing the Canister

The NPWT canister should be changed when full (the alarm will sound), on average every 3 - 5 days, or at least once a week to control odour.

1. Perform hand hygiene, don personal protective equipment.
2. Pause machine; close the clamps on both the canister and the dressing tubing.
3. Disconnect the canister tubing from the dressing tubing.
4. Remove the canister from the unit and dispose.
5. Connect new canister to tubing and place in unit.
6. Open tubing clamps and ensure therapy/suction is resumed.
7. Perform Hand Hygiene

Intermittent Disconnection from NPWT

NPWT should only be disconnected for short periods of time and for not more than a total of two hours per day.

Requirements

- 70% Alcohol swab
- Protective cap or film dressing (e.g. Tegaderm™)
- Personal Protective Equipment

Disconnection

1. Perform hand hygiene, don personal protective equipment

2. Close the clamps on the canister and tubing.
3. Turn the therapy unit off.
4. Disconnect the dressing tubing from the canister tubing.
5. Cover both ends of the tubing with a sterile film dressing (e.g. Tegaderm™).

Reconnection

1. Perform hand hygiene and don personal protective equipment.
2. Remove the film dressing.
3. Wipe both ends of the tubing (dressing and canister) with the 70% Alcohol swab.
4. Reconnect the dressing tubing and the canister tubing.
5. Open both clamps.
6. Turn the therapy unit on. Previous therapy settings will resume.
7. Perform hand hygiene.
8. Dispose of waste.
9. Remove gloves. Perform hand hygiene.

Use of NPWT System At Home

1. If the patient is a suitable candidate for using NPWT at home:
2. Instruct the patient and / or caregiver about how to use the system, potential complications and their signs / symptoms, and what to do if complications occur.
3. Request patient and or caregiver demonstrate use of the system, and document his / her/ their proficiency.
4. Assure the patient and or caregiver understands the machine alarms associated with NPWT system.
5. Provide patient and or caregiver with a copy of NPWT system product information and SCGH hospital's educational pamphlet. Ensure power cord accompanies patient home.
6. Inform HES prior to patient being discharged in case a rental machine is required.
7. Refer to appropriate community health service to continue management.

Cessation

1. As per removal of NPWT dressing
2. Return equipment to HES.

PICO & Prevena Negative Pressure Dressing Application ¹⁶

Requirements

- Personal Protective Equipment (PPE)
- Wet strength bag
- Dressing pack
- Sterile drape pack
- Nonsterile gloves
- Normal Saline
- PICO or Prevena dressing packet

Procedure

1. Administer analgesia if required.
2. Perform hand hygiene.
3. Clean the trolley with a combined detergent/ disinfectant wipe. Allow to dry
4. Gather equipment for the procedure.
5. Perform hand hygiene.
6. Open dressing pack using the corners and add items required using a non-touch technique.
7. Perform hand hygiene and put on non-sterile gloves.

8. Place sterile drape in position.
9. Clean wound base and peri-wound skin.
10. Ensure PICO/Prevena absorbent foam dressing is at least 2 cm bigger in circumference than the wound.
11. Apply the PICO/ Prevena dressing. The port should be situated towards the top or head of the patient.
12. Connect the tubing to the machine.
13. Insert the batteries into the machine.
14. Turn machine on and commence therapy.
15. Check that therapy/suction is working and that there are no leaks.
16. Use supplied additional film to seal any detected leaks.
 - a. Document the therapy settings
 - i. PICO 80mmhg
 - ii. Prevena 125mmhg
17. Dispose of waste.
18. Remove gloves. Perform hand hygiene.
19. Clean the trolley with a combined detergent/ disinfectant wipe.
20. Perform hand hygiene.

Discharge Planning

Where patients require a continuation of management of their wound referral to the appropriate clinical services on discharge should be performed.

- Ensure referral and current wound management plan are supplied to the service provider
- Provide education and a current wound management plan to the patient and/or carer
- Provide additional wound supplies on discharge. Ensure NPWT charger is sent home with patient.
- Ensure the patient/carer are aware of any follow –up appointments and /or service referrals
- Provide education to the patient/carer regarding adverse signs and symptoms:
 - Infection, Wound deterioration,
 - Other potential problems, e.g. pain or bleeding

Wound Management

Monday to Friday: 0730-1530

Call HES on Ext: 71482

Please sign out the device with the following information: Patient details, type of device, serial number or MTP number and ward number. Bring patient stickers x 3

Inpatients and Outpatients
discharged to Homelink/HITH, PDC and
OPH.

SCGH NPWT Device
Ward to collect from HES

SCGH Owned KCI Device

15 KCI ActiVacs (small)
3 KCI UltraVacs (large)

SCGH owned device can be
identified by the yellow MTP label.

Smith and Nephew Lease Device

Renasys Touch

Identified by the serial number on the
back of the device.

Patient Discharged Home

If patient is to be discharged, call or email HES to inform of discharge.
scghequimentservice@health.wa.gov.au

scghequimentservice@health.wa.gov.au please call 71482 or email name,

NPWT ceased

Clinical area to remove canister, discard bag, wipe down and return
device with charger to HES for filter change and quality assurance testing.

DO NOT USE ON ANOTHER PATIENT

Patient Discharge/Transfer to
Silver Chain, another Hospital/Nursing Home.

SCGH NPWT KCI Rental Device
(Office hours)
Ward to collect device from HES

KCI Rental ActiVac

Devices are on top shelf in locked black boxes.
Follow the guidelines to contact KCI to unlock
the box and access the rental device.

Document patient information and asset
number of the device on the sign out sheet.

SRN Process for Rental Units

When patient is transferred to another hospital, post-acute care or
nursing home, rental costs will be transferred to that facility.

Please email the following information to:

scghequimentservice@health.wa.gov.au

- Name of receiving facility
- Name and designation of SRN accepting the rental cost.
- Telephone number

After Hours

Contact After Hours CNS/NM who
can access NPWT devices from
HES.

Theatres can access with their key.

Patient stickers x 3 and information
as required.

Smith and Nephew Lease
Device

Patients can be transferred
with this device

If patient care is transferred
to another facility, follow

HES arranges fleet maintenance, tracking and invoice payment for rental devices
Consumables are available from the Dressing Bank which can be accessed by SRNs and managed by the Wound Management Teams

Wound Management

5.0 Compression Bandaging ¹⁸

5.1 Medical Requirements

Refer to medical staff or wound management team for order for compression bandaging as well as the type and level of compression.

Compression Bandaging Pressures:

- 2 Layer, Light to moderate compression 20-25mmHg;
- 3 Layer Moderate compression 25-35mmHg;
- 4 Layer Moderate to high compression 35-40mmHg;

Safety Information

Criteria for application of compression bandaging include medical orders and application by a competent staff member:

Indications¹⁷	Contraindications¹⁷
• Peripheral Venous Disease • Varicose Veins • Venous Eczema • DVT prophylaxis and treatment • Lymphoedema • Venous Leg Ulcers	• Peripheral Artery Disease • Weak or absent pedal pulses • Ankle Brachial Pressure Index (ABPI's) >0.9 • Toe Pressures >50mmHg

5.2 Procedural Information

Dress all wounds as per Wound Management Care Plan (MR 853)

Ensure bandaging is 'two fingers' below the back of the knee and that you are able to run finger around top of bandaging. Ensure foot flexion.

Light to Moderate Compression: Coban Lite

Requirements:

- Coban Lite-2 Layer
- Securement tape

Procedure:

1. Perform hand hygiene.
2. Gather equipment for the procedure.
3. Perform hand hygiene.
4. Apply layer 1, in a spiral, with 50% overlap, toe to knee.
5. Apply layer 2, in a spiral, toe to knee.
6. Secure with tape.
7. Discard waste.
8. Perform hand hygiene.
9. Document in patients notes/ care plan once application complete.

Moderate to High Compression: Profore

Requirements:

- Kerlix or softban
- Crepe bandage or profore 2
- Lastodur lite

- Profore 4
- Securement tape

Procedure:

1. Perform hand hygiene.
2. Gather equipment for the procedure.
3. Perform hand hygiene.
4. Apply first layer- Kerlix or Softban, in a spiral, with 50% overlap but no stretch from toe to knee.
5. Apply second layer- Crepe or profore 2, in a spiral, with 50% overlap but no stretch from toe to knee.
6. Apply third layer- Lastodure Lite, in a figure of 8, with 50% stretch, from toe to knee
7. Apply forth layer- if required, Profore 4, in a spiral, with 50% overlap but no stretch from toe to knee.
8. Secure with tape.
9. Discard waste.
10. Perform hand hygiene.
11. Document in patients notes/ care plan once application complete.

Discharge Planning:

- Where patients require a continuation of management of their wound referral to the appropriate clinical services on discharge should be performed.
- Ensure referral and current wound management plan are supplied to the service provider
- Provide education and a current wound management plan to the patient and or carer
- Provide additional wound supplies on discharge
- Ensure the patient/carers are aware of any follow –up appointments and /or service referrals
- Provide education to the patient/carers regarding adverse signs and symptoms.

6.0 Moisture Associated Skin Damage Prevention and Management 4, 10, ,14,15, 26

The term moisture-associated skin damage (MASD) delineates a spectrum of injury characterised by the inflammation and erosion (or denudation) of the epidermis resulting from prolonged exposure to various sources of moisture and potential irritants (e.g., urine, stool, perspiration, wound exudate, and ostomy effluent).

Technically, MASD is a type of irritant contact dermatitis, but it is an umbrella term that includes four (4) distinct clinical entities:

- incontinence-associated dermatitis (IAD),
- intertriginous dermatitis (ITD),
- peri wound skin damage, and
- peristomal MASD.²¹

It is a common disorder that may occur, for many reasons, to patients at any stage throughout their admission.

6.1 Medical Requirements

Medication based creams may need to be prescribed by a medical officer or nurse practitioner as indicated.

6.2 Safety Information

Patients known allergies need to be checked prior to the commencement of any creams or wound products.

6.3 Nursing Management

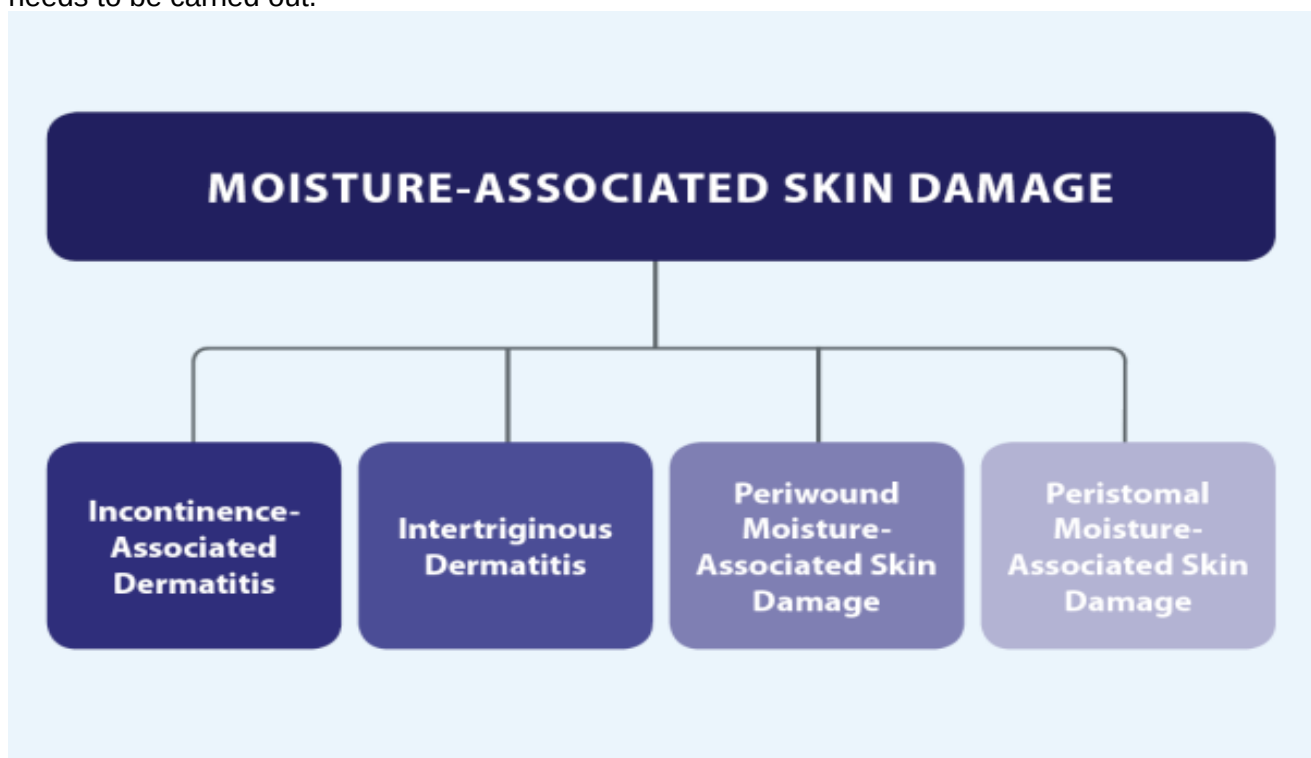
Risk Assessment

Protecting the skin against moisture-associated damage is an important component of comprehensive skin and wound care. A thorough full skin assessment ¹⁰ needs to be conducted to identify if any skin injuries are present and to assess patients' risk.

Intrinsic Factors (5)	Extrinsic Factors (5)
Excessive perspiration	Chemical/biologic irritants
Increased dermal metabolism (i.e., elevated local temperature)	Mechanical stress on the skin (e.g., friction, pressure, shear)
Abnormal skin pH	Fungal / candidiasis proliferation
History of atopy	Seasonal or environmental factors (e.g., humidity)
Deep body folds	Incontinence (urine, faecal, or both) and poor hygiene practices

<u>Risk Factors</u> (5)	<u>Signs & symptoms</u> (5)
Faecal and/or urinary incontinence	Redness-from light pink to dark red
Leakage from stomal output/wound exudate	Inflammation- in patches or large continuous area
Decreased mobility	Warm, firm skin
Excess moisture/perspiration	Lesion
Friction/shear	Pain or tenderness
Previous radiotherapy to the area	Burning
Medications	Itching

Differentiation of moisture associated skin damage from other skin injuries (including pressure injuries) needs to be carried out.



Classification of Moisture Associated Skin Damage ⁽⁵⁾

Category 1a	Mild erythema with no skin loss or signs of infection
Category 1b	Mild erythema with no skin loss but signs of infection present
Category 2a	Severe erythema with skin loss present but no signs of infection
Category 2b	Severe erythema with both skin loss and signs of infection present

Please refer to Table 3, page 23 for images

Prevention

- Prevention and treatment of MASD begin with a thorough assessment of the skin.
- Proper, yet gentle cleansing of the skin should occur as soon as possible after soiling.
- Cleansers should be pH balanced and contain gentle surfactants that protect the lipid profile of the skin.
- No-rinse formulas will allow for cleansing without requiring additional friction.
- Once the skin is cleansed, barrier products should be applied to repel/protect the skin from subsequent episodes of prolonged exposure to moisture and other caustic agents.
- Prevention is effective when the skin is kept clean and dry, a pH neutral soap/skin cleanser is used and barrier cream such e.g., Cavilon is recommended for high-risk patients.

If unsure if your patient has moisture associated dermatitis or a pressure injury, or both, please contact wound management for assessment

Table 3 Illustration of Moisture Associated Skin Damage¹⁴

Category 1: Persistent redness	Category 2: Skin loss
A - Persistent redness without clinical signs of infection  Critical criterion • Persistent redness <i>A variety of tones of redness may be present. Patients with darker skin tones, the skin may be paler or darker than normal, or purple in colour.</i> Additional criteria • Marked areas or discolouration from a previous (healed) skin defect • Shiny appearance of the skin • Macerated skin • Intact vesicles and/or bullae • Skin may feel tense or swollen at palpation • Burning, tingling, itching or pain	2A - Skin loss without clinical signs of infection  Critical criterion • Skin loss <i>Skin loss may present as skin erosion (may result from damaged/eroded vesicles or bullae), denudation or excoriation. The skin damage pattern may be diffuse.</i> Additional criteria • Persistent redness <i>A variety of tones of redness may be present. Patients with darker skin tones, the skin may be paler or darker than normal, or purple in colour</i> • Marked areas or discolouration from a previous (healed) skin defect • Shiny appearance of the skin • Macerated skin • Intact vesicles and/or bullae • Skin may feel tense or swollen at palpation • Burning, tingling, itching or pain
1B - Persistent redness with clinical signs of infection  Critical criteria • Persistent redness <i>A variety of tones of redness may be present. Patients with darker skin tones, the skin may be paler or darker than normal, or purple in colour.</i> • Signs of infection <i>Such as white scaling of the skin (suggesting a fungal infection) or satellite lesions (pustules surrounding the lesion, suggesting a Candida albicans fungal infection).</i> Additional criteria • Marked areas or discolouration from a previous (healed) skin defect • Shiny appearance of the skin • Macerated skin • Intact vesicles and/or bullae • The skin may feel tense or swollen at palpation • Burning, tingling, itching or pain	2B - Skin loss with clinical signs of infection  Critical criteria • Skin loss <i>Skin loss may present as skin erosion (may result from damaged/eroded vesicles or bullae), denudation or excoriation. The skin damage pattern may be diffuse.</i> • Signs of infection <i>Such as white scaling of the skin (suggesting a fungal infection) or satellite lesions (pustules surrounding the lesion, suggesting a Candida albicans fungal infection), slough visible in the wound bed (yellow/brown/greyish), green appearance within the wound bed (suggesting a bacterial infection with Pseudomonas aeruginosa), excessive exudate levels, purulent exudate (pus) or a shiny appearance of the wound bed.</i> Additional criteria • Persistent redness <i>A variety of tones of redness may be present. Patients with darker skin tones, the skin may be paler or darker than normal, or purple in colour</i> • Marked areas or discolouration from a previous (healed) skin defect • Shiny appearance of the skin • Macerated skin • Intact vesicles and/or bullae • Skin may feel tense or swollen at palpation • Burning, tingling, itching or pain

Wound Management

6.6 Management ^{5,14}

Management of moisture associated skin damage is complex and multifactorial. It includes a multidisciplinary team approach and may include specialised equipment, incontinence products, diet, dressing products, skin cleansing agents.

Identify patients that are at risk of Moisture Associated Skin Damage e.g., patients that have urinary and faecal incontinence, obesity, deep skin folds and increased friction with skin-to-skin contact

Prevention

- Daily skin inspections
- TDS and PRN cleansing of the skin with Abeena or water wipes
- Apply a durable barrier cream e.g. Cavilon cream and / or Cavilon continence wipe
- Upgrade the patient's mattress
- Reposition the patient regularly
- Optimise nutrition and hydration
- Optimise the continence toileting regime
- Avoid using occlusive products

Management

- Frequent cleaning of the affected area.
- Gently pat dry the skin -never rub.
- Apply appropriate prescribed creams to the affected area as per the patient's condition e.g., fungal infection, inflammation and / or pain being present.
- Removing the incontinence pad and placing on a bluey if appropriate.
- Use of barrier creams and wipes and pads e.g., Ultrasorb pads in between skin and pannus folds.
- Immediately after an incontinence event, remove soiled clothing or incontinence pads or products.
- Avoid the use of talc, corn-starch-based powders in these areas.

7.0 Wet Wraps

Wet wraps (or wet dressings) are used to increase the penetration of topical medications, increase the skin's moisture content, reduce erythema, promote sleep by relieving discomfort and enhance the removal of skin scales, crusts and exudate by reducing the local skin bacterial burden ⁽¹⁾.

Emollients, steroids and creams are applied to the patients' skin and warmly dampened cotton garments, or towels are applied to the patient for a specific / prescribed time frame.

7.1 Clinical Indications ^{7,22}

- Severe and widespread eczema and dermatitis including atopic dermatitis, discoid dermatitis and contact dermatitis
- Psoriasis
- Drug Eruptions
- Severe Pruritus
- Pityriasis Rubra Pilaris
- Various other widespread inflammatory dermatoses

7.2 Medical Requirement

Topical ointments, creams and steroids should be prescribed by a Medical Doctor or Nurse Practitioner.

7.3 Safety Information

It is imperative that high potency steroids are not applied to the face and other sensitive areas such as the axilla or groin. Patients should be advised that large quantities of paraffin-based products are highly flammable and if the patient is a smoker, they should refrain from leaving the ward for a smoke or should be advised that a thick outer layer of clothing should be worn to decrease chance of ignition.

Ensure that topical ointments and creams for the feet are applied once the patient is resting in bed to decrease falls risk.

7.4 Risk Factors ⁷

As with any medical treatment, there is a possibility of an adverse reaction with Wet Wraps. Nursing staff are required to escalate any abnormal findings to the Medical Team or Shift Co-Ordinator.

Potential adverse reactions to Wet Wraps may include:

- Folliculitis,
- Secondary Infection of the Skin – e.g. Staphylococcus, Pseudomonas, candida, herpes simplex. This may present as purulent exudate, weeping or crusting.
- Skin Atrophy,
- Striae,
- Temperature Dysregulation.

7.5 Procedural Information

Requirements

- PPE as per Standard Precautions / Contact Precautions as applicable to the patient.
- Metal Trolley
- 1 x Large Silver Bucket
- Cut Plastic Sheets
- Clear Plastic Bag
- 6-8 x Bath Towels

- Standard Bed Linen
- Large (Milkshake) Cup
- Creams and Ointment as Prescribed by Medical Officer / Nurse Practitioner.

Preparation

1. Place silver bucket on trolley
2. Line the bucket with clear plastic bag
3. Roll up four (4) towels and place each one in the bucket **vertically**
4. Fill a large milkshake cup with **hot** water and evenly pour over one towel at a time
5. Place two (2) towels roller **horizontally** on top of the wet vertical towels
6. Pour hot water over these (one cup per towel)

Procedure

1. Explain procedure to the patient and obtain consent
2. Ensure privacy and comfort for the patient while resting in bed; close curtains and blinds
3. Place plastic sheets under the patient but on top of the fitted sheet of the bed. This will prevent the linen from becoming wet and will reduce the evaporation of moisture.
4. Perform Hand Hygiene and don PPE
5. Apply ointment and creams as prescribed to affected area. Please note that different topical agents may be applied to different sites of the body
6. Take incontinence pads **off** unless essential
7. Lay one moistened towel on the plastic sheet on the bed and have the patient lay down
8. Apply one warm, moistened towel around each affected limb
9. Cover patient with a second plastic sheet to maintain warmth, contain moisture and topical ointments applied.
10. Doff PPE and ensure patient call bell and belongings are within reach.
11. Leave towels insitu for 20 minutes or as prescribed
12. After prescribed time, **pat** dry the treated skin (using dry towels). Do not rub or wipe the skin as this will cause excess friction and shear to newly moistened and treated skin.
13. Apply post-wrap topical ointments **if** prescribed and perform hand hygiene
14. Advise the patient to dress in light clothing to reduce irritation to the skin
15. Repeat wet dressing procedure TDS or as prescribed
16. Ensure Medication Chart is signed once dressings applied and Inpatient Notes updated accordingly.

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Carer's Recognition Act 2004 (WA).
- Work Health and Safety Act 2020
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018

Policy:

- WA Health Mandatory Policy Consent to Treatment Policy
- NMHS Clinical Documentation Policy
- NMHS Clinical Incident Management policy
- SCGOPHCG Patient Identification
- OPH Procedure Matching Policy
- SCGH Procedure Matching Policy
- SCGOPHCG Infection Prevention and Management Manual - Aseptic Technique Policy
- SCGOPHCG Pressure Injury Risk Management Clinical Practice Guideline
- SCGOPHCG Clinical Photography and Audio-Visual Recordings Policy

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